

PITTSBURNT 2.4

Developer Change Spec

Repository-audited model corrections + staged Crash Test + moving people + readable Before / After + local-first demo

Outcome: No output number is typed into the UI as a showcase value. Every displayed impact must come from a source record, a documented assumption, or a reproducible simulation snapshot.

Decisions locked in this revision

- REMOVE ONLY: Cooling Station / Cooling Node from MVP.
- RESTORE REQUIRED: Shaded Waiting Shelter is not a Cooling Station. Restore shaded_shelter in the schema, UI, map, optimizer and tests.
- FINAL ADJUST SET: exactly two options - Street Tree + Shaded Waiting Shelter.
- MIGRATE: keep all 25 building heights, but attach source, method and confidence metadata to every record.
- CONTEXT: show corridor impact and network-wide 5.2% across 167 km together; never show a bare 5.2%.
- DEPLOY: deterministic local demo and exact-input cached fallback first; cloud only after the core loop is locked.
- ALLOCATE: user enters the real budget; a deterministic optimizer chooses the mix; an LLM only parses intent and explains evidence.
- FILL VISUALLY, NOT COMPUTATIONALLY: surrounding streets, buildings, parks, trees and transit complete the map; only the current demo boundary runs UTCI, agents or Adjust.
- DEMO UX: reveal climate change, UTCI stress, moving people and the vulnerable corridor in sequence; do not return the final red map in one instant.

Product line

"The model does not invent an answer. It tests every feasible city change, spends the entered budget, and explains the winning plan."

1. Change map - exact revisions

Status	Area	2.3 behavior	2.4 requirement
REMOVE	Cooling Station	Cooling Station was correctly targeted for deletion	Delete cooling_station / cooling_node only
RESTORE	Shaded shelter	Deleted together with Cooling Station by mistake	Restore shaded_shelter in schema, UI, map, optimizer and tests
REPLACE	Output payload	Hard-coded 2,840 people and 1,420 minutes	Bind all totals to the completed snapshot; no showcase constants
REPLACE	Population label	exposedPeople could imply observed volume	simulatedAgents unless a documented expansion factor exists
NEW	Budget entry	\$250K hero value	Editable USD field; \$250K is only the initial user input
NEW	Allocation	Greedy ROI recommendation	Re-simulate feasible Tree/Shelter combinations under budget
NEW	AI reasoning	No structured explanation contract	Evidence summary generated from optimizer trace
REPLACE	25 heights	Manual values without provenance	Retain value; append source type, reference, method, confidence
REPLACE	5.2% card	Risk of denominator-free display	5.2% network reduction across 167 km + calculated corridor card
REPLACE	Deployment	Cloud account implied	Local-first, deterministic seed, cached exact-input fallback
FIX	Worker speed	1.20 m/s in brief; 1.30 m/s in spec	Use 1.20 m/s everywhere; label as planning assumption
NEW	Map context	Non-demo area can look empty	Render a muted context shell; keep all simulation corridor-only

Hotfix rule: remove cooling_station / cooling_node only. shaded_shelter is a required MVP intervention and must be restored everywhere. Final allowlist = [tree, shaded_shelter].

2. No fabricated result policy

Every numeric field carries a value status. UI components may display computed results only after snapshot validation succeeds.

Value provenance

Status	Meaning	Allowed UI treatment
source	Directly loaded from a named dataset or record	Show source and retrieval date
computed	Derived by a versioned function from source / assumption inputs	Show formula or calculation tooltip
assumption	Planning input without authoritative local calibration	Show ASSUMPTION badge and allow editing
missing	Required value absent or snapshot hash mismatch	Hide result; show Data not ready

Replace the sample alert payload

```
const snapshot = await runScenario(input);
assert(snapshot.status === 'complete');
assert(snapshot.inputHash === hash(input));

const alertPayload = {
  label: stressBand(snapshot.maxUtciC),
  thresholdC: UTCI_THRESHOLDS.veryStrong,
  populationBasis: snapshot.population.basis,
  simulatedAgents: snapshot.population.sampleSize,
  estimatedPeoplePerHour:
    snapshot.population.expansionFactor == null
      ? null
      : snapshot.population.sampleSize * snapshot.population.expansionFactor,
  severePersonMinutes: snapshot.metrics.severePersonMinutes,
  snapshotId: snapshot.id,
  inputHash: snapshot.inputHash
};
```

UI rule: if expansionFactor is null, display “simulated pedestrians,” never “people per hour.”
The old 2,840 / 1,420 constants are deleted rather than replaced by different mock values.

3. Local-first target architecture

The core demo must work without a cloud account, live API, or language model. Cloud services are optional enhancements after deterministic replay is accepted.

Recommended file map

```
src/
  components/BudgetInput.tsx
  components/AdjustPlanner.tsx
  components/AllocationReasoning.tsx
  components/MapContextShell.tsx
  state/crashTestStore.ts
  lib/valueProvenance.ts
  lib/impactScope.ts
  lib/demoBoundary.ts

simulation/
  exposure_engine.py
  intervention_engine.py      # tree + shaded_shelter only
  candidate_generator.py
  allocation_optimizer.py
  scenario_runner.py

data/
  buildings_25.geojson      # height metadata required
  context/oakland_shell.geojson # display only; simplified + cached
  intervention_costs.json   # source / assumption metadata required
  network_scope.json        # 167-km scope provenance
  cached/default_250k.json  # exact-input fallback only
```

Runtime rule

1. Load local geometry, scenario and cost catalog.
2. Hash all inputs: weather, seed, budget, objective, costs, geometry.
3. Run local simulation and optimizer.
4. If local compute fails, use cache ONLY when `cache.inputHash == inputHash`.
5. Never serve the \$250K cache for a different budget or objective.
6. Add cloud hosting / LLM parsing only after steps 1-5 pass offline.

Deployment acceptance

- The Today -> 2035 -> Adjust -> Re-test loop completes with networking disabled.
- The numeric budget field and manual objective controls work without an LLM.
- A cache hit is visible in debug metadata and exactly matches the current input hash.

4. Data contracts - values, population and impact scope

Core snapshot schema

```
interface ValueMeta {
  status: 'source' | 'computed' | 'assumption' | 'missing';
  sourceLabel?: string;
  sourceUrl?: string;
  method?: string;
  confidence: 'high' | 'medium' | 'low';
}

interface PopulationBasis {
  basis: 'simulated_sample' | 'count_calibrated';
  sampleSize: number; // computed
  expansionFactor: number | null;
  source?: ValueMeta;
}

interface ImpactScope {
  label: string;
  lengthKm: number;
  beforeMetric: number;
  afterMetric: number;
  reductionPct: number; // computed, not typed
  snapshotId: string;
}
```

Calculation contract

```
reduction_pct(before, after) =
  100 * (before - after) / before

corridor.length_km = SUM(unique corridor segment lengths) / 1000
corridor.reduction_pct = reduction_pct(corridor.before, corridor.after)

network.length_km = 167
network.reduction_pct = 5.2

# Release gate for the supplied network result:
assert round(reduction_pct(network.before, network.after), 1) == 5.2
assert network.length_km == 167
assert network.snapshot id and network.dataset hash
```

Critical: “167 km” is the evaluated network extent, not 5.2% of road length. Do not convert 5.2% into kilometres. It is an impact reduction calculated across a 167-km network.

5. Keep 25 building heights - add evidence

Retain the current 25 numeric heights so the demo geometry does not break. Migration adds disclosure; it does not pretend that a legacy manual estimate became measured data.

Required record

```
{
  "building_id": "<existing stable id>",
  "height_m": "<retain existing numeric value>",
  "height_source_type":
    "osm_height | published_floor_count | osm_levels | manual_estimate",
  "height_source_label": "<human-readable source>",
  "height_source_url": "<URL or null>",
  "raw_source_value": "<direct height / floor count / null>",
  "conversion_rule": "direct | levels_x_assumed_floor_height | legacy_manual",
  "assumed_floor_height_m": "<number or null>",
  "confidence": "high | medium | low",
  "verified_at": "<ISO date>",
  "notes": "<limitations>"
}
```

Confidence mapping

Source	Confidence	Treatment
Measured / authoritative direct height	High	Use directly; retain source and accuracy
OSM height with source tag	High / Medium	Use metres after unit validation
Published floor count or OSM building:levels	Medium	Convert with an explicitly stored floor-height assumption
Existing manual prototype estimate	Low	Keep for MVP; tooltip must say Manual estimate

Migration and validation

```
for building in buildings_25:
    if building.has_direct_height_source():
        attach_direct_source_metadata(building)
    elif building.has_published_or_osm_levels():
        attach_level_conversion_metadata(building)
    else:
        building.height_source_type = 'manual_estimate'
        building.conversion_rule = 'legacy_manual'
        building.confidence = 'low'

assert len(buildings_25) == 25
assert all(b.height_source_type and b.confidence for b in buildings_25)
```

OSM documents height, source:height and building:levels; levels exclude underground and roof levels. See source notes on the final page.

6. Exposure and movement - computed, not narrated

The environmental thresholds remain source-backed. Movement defaults remain disclosed planning assumptions until locally calibrated.

Core equations

```
agent_minutes_on_segment = segment.length_m / agent.speed_mps / 60
person_minutes(segment) = SUM(agent_minutes_on_segment)
```

```
UTCI = f(air_temperature, humidity, wind, mean_radiant_temperature)
severe_person_minutes = SUM(person_minutes where UTCI >= 38 C)
heat_load = SUM(max(UTCI - 26 C, 0) * person_minutes)
```

Movement defaults - single source of truth

Persona	speed_mps	Status	UI disclosure
Student	1.30	Planning assumption	Editable sensitivity input
Worker	1.20	Planning assumption	Fixes 2.1 document conflict
Older adult	0.90	Planning assumption	Not a physiological-risk multiplier
Mobility-constrained	0.80	Planning assumption	Use accessible-network filter

Population output rule

- A 100-500 agent run reports simulated agents and simulated person-minutes.
- A real people/hour estimate requires a stored observed count, count period, expansion method and expansion factor.
- Before and After reuse the same seed agents routes weather and time. Only intervention state may change

Red threshold: UTCI >= 38 C is “Very Strong Heat Stress.” This is a thermal-stress class, not a diagnosis or heatstroke probability.

7. Adjust MVP - Tree + Shaded Shelter only

IMPLEMENTATION HOTFIX: Restore shaded_shelter. It is a passive shade intervention at a validated transit waiting node. It is not the removed Cooling Station / Cooling Node and it does not model active cooling, staffed refuge or time spent inside a cooling facility.

Final candidate types

Option	Simulation change	Impact horizon	Site guardrail
Street Tree	Canopy polygon changes shade overlap -> Tmrt -> UTCI	Year 1 and At Maturity	No existing canopy; feasible tree pit; no utility conflict
Shaded Waiting Shelter	Shelter footprint changes WAIT_NODE shade -> Tmrt -> UTCI	Immediate	Transit stop only; no existing / committed shelter; accessible clearance

Intervention code

```
InterventionType = Literal['tree', 'shaded_shelter']

def apply_intervention(world, intervention, horizon):
    next_world = deepcopy(world)
    if intervention.type == 'tree':
        add_tree_canopy(next_world, intervention.geometry, horizon)
    elif intervention.type == 'shaded_shelter':
        shade_wait_node(next_world, intervention.stop_id,
                        intervention.footprint)
    return recompute_tmrt_utci_and_exposure(next_world)

# Required in 2.4 MVP:
assert set(allowed_intervention_types) == {'tree', 'shaded_shelter'}
assert 'shaded_shelter' in candidate_generator.types
assert 'shaded_shelter' in optimizer.allowed_types

# Forbidden anywhere in 2.4 MVP:
assert 'cooling_station' not in allowed_intervention_types
assert 'cooling_node' not in allowed_intervention_types
```

Cost contract

- The optimizer reads unit / site costs from intervention_costs.json; it never embeds \$8K or \$35K in code.
- Every cost includes jurisdiction, scope inclusions, source URL, year and confidence. Comparable bids are proxies, not Pittsburgh quotes.
- The budget owner may replace planning estimates with actual quotes without changing optimizer code.

Pittsburgh alignment

- Oakland policy supports tree canopy, shaded public space and rider amenities.
- DOMI shelter criteria include ridership, weather exposure, community amenities and equity.
- Committed PRTX shelters must be baseline protection in future scenarios, not new Adjust candidates.

8. Real-budget allocation engine

The user can enter \$250K, \$250,000 or 0.25M. The normalized value is 250,000 USD. The same engine must also work for any other valid budget.

Input contract

```
interface AllocationRequest {
  budgetUsd: number;           // normalized user input
  scenarioSnapshotId: string;
  horizon: 'immediate' | 'mature_canopy';
  objectivePopulation: PersonaId | 'all';
  allowedInterventions: ['tree', 'shaded_shelter'];
  costCatalogVersion: string;
  naturalLanguageRequest?: string;
}
```

Deterministic selection objective

Primary: maximize severe_person_minutes_avoided
Secondary: maximize cumulative_heat_load_avoided
Tertiary: minimize total_cost_usd

Constraints:

- total_cost_usd <= budget_usd
- site_feasible(candidate) == true
- no duplicate existing / committed protection
- no overlapping shade double-count
- objective population follows the parsed user request

Combination search

```
def optimize(snapshot, request, candidates):
    beam = [empty_plan(snapshot)]
    while expandable(beam):
        proposals = []
        for plan in beam:
            for candidate in feasible_remaining(plan, candidates, request):
                if plan.cost + candidate.cost <= request.budget_usd:
                    after = simulate(snapshot, plan.items + [candidate], request.horizon)
                    proposals.append(score(plan + candidate, after))
        beam = deterministic_top_k(proposals, tie_break='candidate_id')
    return best_lexicographic(beam)
```

Claim language: call the result “simulation-recommended allocation.” Call it a global optimum only if exhaustive / exact optimization is implemented and tested.

9. AI reasoning without invented numbers

The LLM is an interface and explanation layer. The simulator and deterministic optimizer own all quantities and choices.

Natural-language flow

User:
"Use \$250K. Prioritize older adults near Fifth Avenue bus stops."

LLM / parser output:
budgetUsd = 250000
objectivePopulation = 'older_adult'
corridor = 'Fifth Ave'
allowedInterventions = ['tree', 'shaded_shelter']

Optimizer output:
selected candidate IDs, exact costs, before / after snapshots,
marginal benefits, rejected candidates and constraint reasons

LLM explanation input:
optimizer trace only - no access to invent costs or impacts

Reasoning card fields

Why this allocation

Budget entered:	{request.budgetUsd}
Trees selected:	{plan.countByType.tree}
Shelters selected:	{plan.countByType.shaded_shelter}
Spend / remaining:	{plan.totalCostUsd} / {plan.remainingUsd}
Severe minutes saved:	{plan.deltaSeverePersonMinutes}
Heat load reduced:	{plan.deltaHeatLoad}
Highest-value site:	{plan.rankTrace[0].candidateId}
Reason:	generated only from rankTrace evidence
Cost basis:	{plan.costCatalogVersion}
Confidence:	{plan.minimumInputConfidence}

Guardrails

- Show the parsed budget, geography, population and allowed interventions for confirmation before optimization.
- If the LLM is unavailable, keep the numeric input, dropdowns and optimizer fully functional.
- Do not expose chain-of-thought. Show a concise evidence summary: cost, impacted minutes, feasibility and why alternatives ranked lower.
- Never let the LLM modify cost, shade, UTCI or population fields in the validated snapshot.

10. Impact display - corridor and 167-km network

Both scopes appear in the same result panel so the dramatic corridor improvement is not confused with city-network impact.

Required card layout

```
SELECTED CORRIDORS
{corridor.reductionPct}% reduction
across {corridor.lengthKm} km
{corridor.beforeMetric} -> {corridor.afterMetric}
Snapshot: {corridor.snapshotId}
```

```
167-KM NETWORK CONTEXT
5.2% reduction across 167 km
{network.beforeMetric} -> {network.afterMetric}
Snapshot: {network.snapshotId}
Dataset: {network.datasetHash}
```

Never render: "5.2% IMPROVEMENT" by itself.

Release gates

- Corridor percentage, length and before / after values are calculated from the selected corridor snapshot.
- The 5.2% card renders only when the 167-km cached evaluation contains before, after, snapshot ID and dataset hash.
- If corridor and network metrics use different objectives or time horizons, label both and do not compare them directly.

Visual sequence

Beat	Implementation	Impact
1	Climate rail recomputes Tmrt / UTCI; locked cards stay fixed	Threshold crossing feels physical, not staged
2	Environment-only rank -> Add People rank	Hottest street can differ from highest human exposure
3	Budget visibly allocates to selected candidate IDs	AI plan feels like a decision, not a filter
4	Tree canopy grows by horizon; shelter shade appears immediately	Time-to-benefit becomes legible
5	Synchronized same-agent Before / After replay	Red trails shrink under controlled comparison
6	Corridor card + 167-km context card land together	Large local effect stays honest at network scale

11. Visual completeness without scope creep

The overview must feel like a finished city product, not an isolated strip floating in an empty map. Surrounding geography is rendered as a context shell; the current Forbes / Fifth / Craig-Centre demo geometry remains the only live simulation area.

Layer and interaction contract

Layer	Outside demo boundary	Inside live boundary
Roads	Primary + minor streets, muted neutral styling	Full contrast; eligible segment IDs highlighted
Buildings	Footprints / light extrusion for urban texture only	25 source-labelled height estimates drive shadow model
Landscape	Parks, river, campus, canopy texture at low saturation	Tree geometry may affect shade when validated
Transit	Routes, stops and labels as static reference	Validated stops may become shelter candidates
Heat + people	No UTCI bands, agents, trails or exposure metrics	Crash Test renders UTCI, agents and exposure trails
Adjust	Disabled; click reads "Context only - not simulated"	Tree + Shaded Shelter candidates are selectable

Visual treatment

- Match the same basemap, lighting direction, line weights and typography across both scopes so there is no gray or white moat.
- Use roughly 25-45% opacity and reduced saturation outside the live boundary; keep the demo corridors at full opacity and contrast.
- Draw a subtle focus mask or boundary label: LIVE SIMULATION AREA. Fade at the edge instead of creating a hard rectangular crop.
- Allow normal pan / zoom over context. When Crash Test starts, ease the camera back to the live boundary before the climate animation.

Scope gate - required in every compute path

```
type LayerScope = 'context' | 'live_simulation';

const isDemoSegment = (segmentId: string) =>
  demoBoundary.segmentIds.has(segmentId);

const eligibleSegments = allSegments.filter(s => isDemoSegment(s.id));
const contextSegments = allSegments.filter(s => !isDemoSegment(s.id));

runUtci(eligibleSegments);           // never contextSegments
runAgents(eligibleSegments);        // never contextSegments
generateCandidates(eligibleSegments);

assert(results.every(r => isDemoSegment(r.segmentId)));
assert(interventions.every(i => isDemoSegment(i.segmentId)));
```

Performance and honesty gates

- Load the context shell from simplified cached GeoJSON or vector tiles; it must not trigger city-wide routing, shadow or exposure computation.
- At overview zoom, the map is visually populated. During Crash Test and Adjust, only the live area changes state.
- A click outside the boundary never returns a hotspot rank, person-minutes, recommendation or fabricated building-height confidence.

12. Repository audit - required corrections

The uploaded implementation was reviewed against this brief. The frontend production build succeeds after dependency installation and all 13 existing engine property tests pass, but those tests validate the old Heat Index model. They do not validate the approved UTCI model, corridor-only scope, shelter waiting exposure or staged demo experience.

Priority	Current repository	Required 2.4 correction
P0	Heat Index + 8 C x sun exposure in api/engine.py	Migrate to Tmrt -> UTCI; add pedestrian wind and solar inputs
P0	Red = relative p95 exposure ramp	Thermal red = UTCI >= 38 C; human hotspot is a separate overlay
P0	"Today" = 2005-2024 hot-day median, not live weather	Rename "Observed hot-day baseline"; never present it as current weather
P0	2050 = trend extrapolation evaluated at 2062	Replace with CMIP6-LOCA2 2050-2074 window; no extrapolated hero value
P0	All 2,409 segments / 167.16 km are scored and optimized	Live gate = Forbes + Fifth + South Craig only: 156 segments / 11.41 km
P0	shade_structure shades 15 m of walking path	Rename shaded_shelter and model validated transit WAIT_NODE exposure
P0	Primary score multiplies planning weights 1.4 / 1.6	Primary exposure stays unweighted; vulnerability is a filter / planning priority
P0	"All pedestrians" is another 300-trip sample	All = aggregation of persona-tagged agents, not an independently sampled cohort
P1	Budget is five preset buttons only	Editable USD input + validation; presets may remain as shortcuts
P1	Trips are hidden static purple lines	Show 40-60 deterministic moving agents with visible exposure state
P1	25 candidate height entries are null; output lacks confidence	Keep resolved estimates but add source, method, reference and confidence
P1	npm ci fails; ESLint reports prefer-const; Makefile assumes .venv	Regenerate lockfile, fix lint, add reproducible environment setup target

Current default snapshot - audit evidence only: 2035 / 3 PM / Older adults currently returns air temperature 34.34 C, Heat Index 35.76 C, 2,082.32 -> 1,974.35 weighted score, 5.185% across 167.16 km, and 208 trees + 0 shade structures for \$249,600. Shelter-only reduces the old score by 0.768%. These are reproducible current-code outputs, not final showcase values; recompute after UTCI, scope and waiting-exposure migration.

13. Crash Test experience - staged, not instant

A fast engine is good; an instantaneous reveal is not a compelling demo. Fetch and validate one snapshot, then reveal its real values through a controlled 5-6 second narrative. The animation presents completed computation; it must never fabricate intermediate numbers.

Required sequence after RUN CRASH TEST

Time	Stage	What changes on screen	Data shown
0.0-0.4 s	Lock	Button becomes RUNNING; camera eases to live boundary	Scenario, hour, persona, snapshot ID
0.4-1.6 s	1. Climate rises	Year rail moves Baseline -> 2035 / 2050; counters tick upward; ambient map warms	Air temp, RH, pedestrian wind, solar input
1.6-2.8 s	2. Thermal stress	Tmrt and UTCI count up; stress chip changes band; only live corridors receive a thermal tint	UTCI value + standard stress class
2.8-4.8 s	3. People exposed	Representative walkers begin moving; icons warm and leave short exposure trails; counter accumulates	Walking + waiting severe person-minutes
4.8-6.0 s	4. Vulnerable street	Camera settles on top corridor; hotspot halo pulses; WHY THIS STREET card lands	Heat + shade + people + duration contributors

Demo orchestration state

```
type DemoStage =  
  | 'idle' | 'loading' | 'climate' | 'thermal'  
  | 'people' | 'hotspot' | 'complete' | 'adapting' | 'replay';  
  
const snapshot = await crashTest(selection);  
validateSnapshot(snapshot, selection);  
  
await reveal('climate', 1200, snapshot.weather);  
await reveal('thermal', 1200, snapshot.thermal);  
await reveal('people', 2000, snapshot.agentStates);  
await reveal('hotspot', 1200, snapshot.hotspots);  
setStage('complete');
```

Interaction and accessibility rules

- Keep Skip animation and Replay; a second run with identical inputs must replay identically.
- Disable input changes only during the sequence; allow Reset at all times. Respect reduced-motion settings with cross-fades and counters without map travel.
- The final panel appears only after hotspot stage. Do not show “Calculating” after the validated snapshot has already returned.
- Recommended pacing is 5-6 seconds. Longer than 8 seconds risks feeling slow during judging.

14. Demo UI hierarchy, people and Adjust replay

The current 310 px sidebar, 330 px result card and repeated 10-12 px labels read like an admin tool. Rebuild the hero state for projection-distance viewing: fewer controls visible at once, larger type, and a three-layer result story.

Information hierarchy

Layer	Primary UI	Minimum display treatment
1. Weather input	Air temperature, humidity, wind, solar / shade	14-16 px label; 24-32 px value; show delta from baseline
2. Thermal stress	UTCI + Moderate / Strong / Very Strong / Extreme	32-48 px UTCI; explicit band; red only from 38 C
3. Human exposure	Severe Heat Exposure Minutes	36-48 px total; split Walking and Waiting beneath
4. Explanation	Why this corridor?	Three short drivers: thermal stress, people, duration / wait

Moving people - required wow layer

- Display 40-60 deterministic representative agents sampled from the active persona cohort. Label “60 of N simulated agents shown”; never imply observed footfall.
- Animate each agent along its actual route geometry. Color is derived from the current segment UTCI band; a growing halo / short trail represents that agent's cumulative exposure.
- At hotspots, cluster markers may stack visually into a pulsing count badge. The count must be simulated agents currently present, not a decorative random number.
- Keep streets mostly neutral. Use a restrained UTCI tint plus agent trails / hotspot halos so the whole network does not become one red block.

Tree + Shaded Waiting Shelter objective

```
walking_severe_minutes = SUM(
  agent_minutes_on_segment WHERE segment_utci_c >= 38)

waiting_severe_minutes = SUM(
  stop_arrivals * mean_wait_minutes WHERE stop_utci_c >= 38)

total_severe_minutes = walking_severe_minutes + waiting_severe_minutes
optimizer value = (before total - after total) / cost usd
```

Data rule: use validated stop locations and available ridership / boarding data. If mean wait or arrival volume is unavailable, use an editable planning estimate with source status and sensitivity range. Do not silently invent a wait time.

ADAPT visual replay

- After optimization, animate selected tree and shelter icons onto exact candidate IDs, reveal spend by type, then replay the same agents on the same timeline.
- Result panel shows environmental and human change together: UTCI before -> after, Walking severe minutes, Waiting severe minutes, total reduction and cost.
- Use ghosted Before trails behind the After replay so the reduction is visible without relying on one percentage.

15. Adjust choreography - show cause before effect

ADJUST must visualize the causal chain: selected site -> intervention appears -> shade footprint appears -> local UTCI changes -> the same people accumulate less exposure. Do not jump from a budget input directly to one final percentage.

Recommended 8-second sequence

Time	Stage	What the judge sees	Bound data
0.0-0.6 s	Lock plan	Confirm budget, objective and live geography; freeze the Before snapshot	Input hash, snapshot ID, budget
0.6-1.4 s	Rank sites	Real candidate points pulse; selected sites resolve in optimizer order	Candidate ID, marginal benefit / cost
1.4-3.0 s	Place	Tree icons pop; shelter icons land; budget bar ticks down from the trace	Exact point, type, cost, running spend
3.0-4.2 s	Grow shade	Tree canopy blooms with horizon label; shelter roof footprint unfolds	Shade footprint, Year 1 / At maturity
4.2-5.4 s	Cool locally	Only affected geometry interpolates from Before UTCI to After UTCI	Before / After UTCI, affected segment or stop
5.4-7.4 s	Re-test	The same agents walk and wait again; ghosted Before trails remain behind	Same agent IDs, routes, waits and clock
7.4-8.0 s	Land result	Spend, mix and Walking / Waiting / Total saved cards settle together	Optimizer trace and exposure deltas

Placement output contract

```
type Placement = {
  placementId: string; candidateId: string;
  kind: 'tree' | 'shaded_shelter';
  segmentId?: string; stopId?: string;
  point: GeoJSON.Point; shadeFootprint: GeoJSON.Polygon;
  costUsd: number; beforeUtcic: number; afterUtcic: number;
  walkingMinutesSaved: number; waitingMinutesSaved: number;
  confidence: 'high' | 'medium' | 'low';
};
```

Animation and map-layer rules

- Tree pop: scale 0.25 -> 1.15 -> 1 with 120 ms stagger. Shelter pop: 0.25 -> 1.12 -> 1 with 220 ms stagger. Use distinct tree-green and shelter-teal shapes, not color alone.
- Animate the first 12 placements individually; reveal the remainder in corridor batches. Never make the audience wait through 208 identical pops.
- An icon appears only at its exact point. Its shade footprint follows the icon. Cooling follows the footprint. A shelter cannot cool an entire street unless the computed footprint supports it.
- Layer order: context -> roads -> Before thermal tint -> placement points -> shade footprints -> After thermal tint -> agents -> labels.

Readability and trust

- Pin one large stage headline above the map: 1 Siting solutions, 2 Adding shade, 3 Cooling locations, 4 Re-testing the same people.
- Keep a persistent budget bar with Spent / Remaining and live Tree / Shelter counts. Clicking an icon reveals why here, cost, Before / After UTCI, walking / waiting minutes saved and confidence.
- If the optimizer selects zero shelters, say why using the trace; never add a decorative shelter. Always show Tree cooling horizon as Year 1 or At maturity.
- Provide Replay, Skip and reduced-motion cross-fades. Numbers must finish on the exact snapshot values, even when animation frames are dropped.

16. Acceptance tests and implementation order

Acceptance tests

- A clean checkout passes `npm ci`, lint, production build and documented Python setup; no uncommitted lockfile repair is required.
- Every displayed value resolves to the active snapshot ID and input hash; no showcase constant such as 2840 or 1420 remains.
- Weather payload includes air temperature, RH, pedestrian-level wind and solar input; segment payload includes Tmrt and UTCI.
- Thermal red begins at UTCI 38 C. A relative percentile may rank human hotspots, but cannot redefine the thermal-stress color.
- “Observed hot-day baseline” replaces “Today” unless a real current-weather service is actually connected and time-stamped.
- Only the 156 segments / 11.41 km on Forbes, Fifth and South Craig enter UTCI, agents, ranking or Adjust; context remains display-only.
- Primary walking, waiting and total severe minutes are unweighted. Planning priority is stored and shown separately.
- All pedestrians aggregates persona-tagged agents; it is not another independently generated 300-trip population.
- The map visibly animates 40-60 deterministic agents and labels the displayed sample; repeated runs reproduce the same motion.
- Exactly two interventions exist: `tree` and `shaded_shelter`. `Cooling Station` and `shade_structure` are absent.
- Walking severe minutes + waiting severe minutes = total severe minutes before and after every intervention.
- A typed budget changes the feasible plan; selected cost never exceeds it; the explanation uses only optimizer-trace values.
- The 5-6 second Crash Test follows `climate -> thermal -> people -> hotspot`, supports `Replay / Skip` and respects reduced motion.
- Adjust renders exact placement coordinates in optimizer order, then shade, local UTCI change and the same-agent replay; every icon, cost and effect resolves to one placement record.
- The network card always reads “5.2% reduction across 167 km” with provenance; no bare 5.2% appears.
- All 25 designated building estimates include source type, method, reference and confidence metadata.

Build order

Order	Task	Exit criterion
1	Repair clean-build path	<code>npm ci</code> , lint, build and Python setup pass from fresh checkout
2	Migrate Heat Index engine to Tmrt / UTCI	Payload and tests use standard UTCI bands; LOCA2 windows replace extrapolation
3	Gate live scope and repair cohorts	Only 156 hero segments compute; All aggregates persona-tagged agents
4	Restore shelter + add waiting exposure	Allowlist is <code>tree</code> + <code>shaded_shelter</code> ; walking / waiting metrics reconcile
5	Add editable budget, provenance and optimizer trace	Any valid budget produces a reproducible feasible plan
6	Build staged Crash Test, Adjust choreography and moving agents	Cause-first placement replay is deterministic; first 12 placements animate individually
7	Rebuild result hierarchy and impact scopes	Readable weather / UTCI / exposure; 5.2% cannot appear without 167 km
8	Lock offline cache, then optional cloud	Manual controls and exact-input fallback survive API / LLM failure

Definition of done

A judge watches a baseline hot day become a future heatwave, sees UTCI cross a standard stress threshold, follows the same simulated people as exposure accumulates, enters a budget, receives a Tree + Shaded Waiting Shelter plan, and watches the same agents replay with lower walking and waiting exposure - with every value traceable to the active snapshot.

17. Sources and data-status notes

Source-backed rules

- UTCI: standard inputs and heat-stress bands, including Very Strong Heat Stress from 38 C. [Copernicus UTCI](#)
- Climate windows: CMIP6-LOCA2 27-model release and 1981-2010, 2025-2049, 2050-2074 periods. [USGS LOCA2 release](#)
- Oakland policy: tree canopy, shaded parks and rider amenities including shade. [Oakland Plan](#)
- Shelter placement: DOMI criteria include average daily ons, weather exposure, community amenities and equity. [Transit Stop Improvement Program](#)
- Committed future baseline: PRTX University Line includes 23 new shelters and is scheduled through 2027. [PRTX University Line](#)
- Building height metadata: OSM documents direct height, source:height and building:levels. [OSM height](#) / [OSM building:levels](#)

Cost evidence - use as labelled comparators only

- 2024 municipal street-tree bid examples list direct planting prices; they do not establish Pittsburgh engineered tree-pit cost. [Creve Coeur bid tabulation](#)
- 2025 public shade-structure bid examples provide a magnitude check; they are not Pittsburgh transit-shelter quotes. [Charlotte County bid tabulation](#)

Project-provided values requiring internal artifact binding

Value	Current status	Release requirement
25 designated heights	Candidate file has 25 null values; resolved rows use modeled_default and lack confidence	Retain resolved estimates; attach record-level method, reference and confidence
5.2% across 167 km	Current old-engine result is 5.185% across 167.16 km	Keep only as labelled network context; recompute after UTCI migration
\$250K	Current default and one of five preset buttons	Make editable; normalize to 250,000 USD; recompute and hash exact input
Shelter wait inputs	No stop arrivals, wait duration or WAIT_NODE exposure in repository	Add source-labelled ridership / GTFS inputs or editable planning assumptions

Repository status: the uploaded build contains 2,409 scored segments / 167.16 km, 1,493 generated trips across five independently sampled populations, a Heat Index engine, and static fallback results. The implementation audit above is based on those artifacts. It does not treat old-engine outputs as final UTCI results or invent missing shelter-wait, wind, Tmrt or confidence data.

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